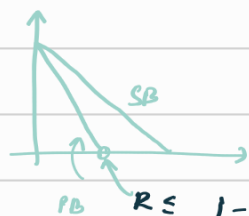


Maximal Distance Separable

Singleton Bound

$$d \leq n - k + 1$$



$$R \leq 1 - \frac{a}{q-1}$$

Reed Solomon Code

$q \geq n$, RS code is MDS

$$G = \begin{bmatrix} \text{K} & \text{n-k} \\ \text{n} & \end{bmatrix} \quad H = \begin{bmatrix} \text{n-k} & \text{d} \\ \text{n} & \end{bmatrix}$$

$d = n - k + 1$
 $d \geq 1$

MDS conjecture: For almost all k , a k -dim code is MDS only if $n \leq q-1$

MDS code needs $q \geq n - d + 2$

If: MDS code (n, k) $d = n - k + 1$. Puncture at $(d-3)$ positions

↳ Resulting code $(n - d + 3, k, = 3)$ $\left\{ \begin{array}{l} \text{Puncturing} \Rightarrow \geq 3 \\ \text{Singleton} \Rightarrow \leq 3 \end{array} \right\} \Rightarrow = 3$

sphere packing: $q^{\frac{n-d+1}{2}} \leq \frac{q^{n-d+3}}{1 + (q-1)(n-d+3)}$

$$q \geq n - d + 2$$

Reed Solomon Code

Polynomials over $\mathbb{F}_q$ $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_dx^d$

Properties:

$\in \mathbb{F}_q$

→ $f(x)$ with degree d over $\mathbb{F}_q$ has at most d roots.

Ex $\mathbb{F}_3$ $f(x) = x^2 - 1 \rightarrow \{0, 2\}$ (roots)

$f(x) = x^2 + 1 \rightarrow \emptyset$

$f(x) = x^2 + 2x + 1 \rightarrow \{2\}$

(Evaluating the function)

$$\begin{array}{ccc} 0 & 1 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 2 \\ 1 & 1 & 0 \end{array}$$

$$(x - r_i) \mid f(x)$$

$$\vdots$$

$$(x - r_d) \mid f(x)$$

$$\Rightarrow \text{d+1 polynomial} \mid f(x)$$

contradiction

→ 'Interpolation'

Given $\{(x_i, y_i)\}_{i=1}^{d+1}$ $\{ \text{distinct } x_i \}$ s.t. $f(x_i) = y_i$

$\Rightarrow \exists!$ polynomial f with degree $\leq d$ over $\mathbb{F}_q$

s.t. $f(x_i) = y_i$

Pf: $f(x) = a_0 + a_1x + \dots + a_dx^d$

$$\bar{x} a = y$$

$$\begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^d \\ 1 & x_2 & \dots & x_2^d \\ \vdots & & & & \\ 1 & \dots & x_{d+1}^d \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_d \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{d+1} \end{bmatrix}$$

Full rank matrix $\Rightarrow a = \bar{x}^{-1}y$
Vander monde matrix

Lagrange Interpolation

$$p(x) = \sum_{i=1}^{d+1} f(x_i) \cdot l_i(x)$$

$$l_i(x) = \prod_{j \neq i} \frac{(x - x_j)}{(x_i - x_j)} \quad \begin{cases} 0 & \text{for } x = x_j, j \neq i \\ 1 & \text{for } x = x_i \end{cases}$$

Polynomial Interpretation of RS codes

$q > n \geq k$ RS code
for evaluation pts $\bar{r} = (r_1, r_2, \dots, r_n)$ taking non zero pts.

$$RS(\bar{r}, n, k) = \left\{ (f(r_1), f(r_2), \dots, f(r_n)) : f(x) \in \mathbb{F}_q[x], \deg(f) \leq k-1 \right\}$$

q^k polynomials
Message is polynomial

Message bits $\{u_0, u_1, \dots, u_{k-1}\}$

$$f_u(x) = u_0 + u_1x + u_2x^2 + \dots + u_{k-1}x^{k-1}$$

Evaluated at $\bar{r}$

Code word $(f(r_1), f(r_2), \dots, f(r_n))$

Same at max $k-1$ positions
 $\Rightarrow d \geq n - (k-1) = n - k + 1$

Claim: $RS(\bar{r}, n, k)$ is MDS

$$d = n - k + 1$$

no. of zeros is $\leq k-1 \Rightarrow$ non zero entries $\geq n - k + 1$

Generator matrix:

$$[u_0 \ u_1 \ \dots \ u_{k-1}] \underbrace{\begin{bmatrix} 1 & 1 & \dots & 1 \\ r_1 & r_2 & \dots & r_n \\ \vdots & & & \vdots \\ r_1^{k-1} & r_2^{k-1} & \dots & r_n^{k-1} \end{bmatrix}}_G = C$$

Even if we get just k code bits, we can reconstruct the message bits.
 → Dual of MDS is also an MDS code.

Dual View

$\mathbb{F}_q, \mathbb{F}_q^* \rightarrow$ multiplicative groups

consisting of all nonzero elements from $\mathbb{F}_q$.

$\mathbb{F}_3, \mathbb{F}_3^* = \{1, 2\}$ under mod 3 multiplication

Fact: $\mathbb{F}_q^*$ is cyclic, i.e., $\exists \alpha \in \mathbb{F}_q^*$ which generates $\mathbb{F}_q^*$
 $\alpha^{q-1} = 1$

$$\langle 1 \rangle = \{1\}$$

$$\langle 4 \rangle = \{1, 4\}$$

$$\langle 2 \rangle = \langle 3 \rangle = \{1, 2, 3, 4\}$$

Fact 2: For any $0 < l < q-1$

$$\sum_{r \in \mathbb{F}_q} r^l = 0$$

Pf:

$$\sum r^l = \sum_{j=0}^{q-2} \alpha^{jl} = \frac{1 - (\alpha^l)^{q-1}}{1 - \alpha^l}$$

$$\alpha^{q-1} = 1$$

α is called a primitive ^{element} of $\mathbb{F}_q$

$$\bar{\alpha} = \{\alpha^0, \alpha^1, \dots, \alpha^{q-1}\}$$

$$RS(\bar{\alpha}, n, k) =$$

$$(c_0, c_1, \dots, c_{n-1}) \in \mathbb{F}_q^n$$

$$\text{s.t. } C(x) = c_0 + c_1 x + \dots + c_{n-1} x^{n-1}$$

$$\& C(\alpha^j) = 0 \quad \forall j \in 1, 2, \dots, n-k$$

$$\begin{matrix} n-k \\ \left\{ \begin{array}{c} \begin{matrix} \begin{matrix} 1 & \alpha & \alpha^2 & \dots & \alpha^{n-1} \\ 1 & \alpha^2 & (\alpha^2)^2 & \dots & (\alpha^2)^{n-1} \\ \vdots & & & & \\ 1 & \alpha^{n-k} & (\alpha^{n-k})^2 & \dots & (\alpha^{n-k})^{n-1} \end{matrix} \\ (n-k) \times n \end{matrix} \end{array} \right\} \begin{bmatrix} c_0 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \vdots \\ 0 \\ \vdots \\ 1 \end{bmatrix} \end{matrix}$$

$$G = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \alpha & \alpha^2 & & \alpha^{n-1} \\ \vdots & \alpha^2 & (\alpha^2)^2 & & \\ 1 & & & & (\alpha^{n-1})^{k-1} \end{bmatrix}$$

$n \quad k \times n$

Review in moodle notes.

Decoding of Reed Solomon Codes

Berlekamp - Massey order ($n \log n$)

Berlekamp - Welch - order (n^3)

decode RS ($\mathbb{F}, n, k$) upto e errors

$$e \leq \left\lfloor \frac{d_{\min}-1}{2} \right\rfloor = \left\lfloor \frac{n-k}{2} \right\rfloor$$

$$r = (r_1, \dots, r_n)$$

Find a polynomial $f \in \mathbb{F}_q[x]$

- ① $\deg(f) \leq k-1$ ② $f(r_i) \neq r_i$ in at most e locations

Idea : Error locator polynomial

$$E(x) = \prod_{i: r_i \neq f(r_i)} (x - r_i)$$

$$\deg(E(x)) \leq e$$



Don't know it before decoding.

$$\text{But : } \forall i, \quad r_i E(r_i) = \underbrace{f(r_i) E(r_i)}_{Q(r_i)}$$

$$\text{Find } E(x), Q(x), \quad \hat{f}(x) = \frac{Q(x)}{E(x)}$$

① Find:

(a) Monic polynomial $E(x)$ with degree e .

(b) a polynomial $Q(x)$ of degree $\leq e+k-1$
s.t. $r_i E(r_i) = Q(r_i) \quad \forall i$

If unable to find E, Q , then declare error

$$\text{② } \hat{f}(x) = \frac{Q(x)}{E(x)}$$

If $\hat{f}(r_i) \neq r_i$ at more than e locations,
declare error o.w. o/p $\hat{f}$

Ⓐ How do we know that we will find some E, Q ?

Ⓑ How do we know that we will get a consistent $\hat{f}$?

Claim 1: If $\exists f$ with $\deg(f) \leq k-1$

Existence

s.t. $f(r_i) \neq r_i$ different in $\leq e$ locations,
then $\exists E \in \mathbb{Q}$ that satisfy

$$\text{Pf: } E(x) = \prod_{i: f(r_i) \neq r_i} (x - r_i)^{\Delta_{fr}}$$

$\uparrow$
degree = e

Δ_{fr} is the no. of places where $f(r_i) \neq r_i$

$$Q(x) = E(x) \cdot f(x)$$

Claim 2: If $(Q_1, E_1) \neq (Q_2, E_2)$ satisfy $(*)$, then $\frac{Q_1}{E_1} = \frac{Q_2}{E_2}$

Consistency

$$\text{Pf: } R(x) = \underbrace{Q_1(x)}_{e+k-1} \underbrace{E_2(x)}_e - Q_2(x) E_1(x)$$

$$\deg R(x) \leq 2e + k - 1 = 2 \left\lfloor \frac{n-k}{2} \right\rfloor + k - 1 < n$$

$$\forall i, R(r_i) = \underbrace{Q_1(r_i)}_{r_i E_1(r_i)} E_2(r_i) - \underbrace{Q_2(r_i)}_{r_i E_2(r_i)} E_1(r_i) = 0$$

$\leftarrow n$ roots $\deg R < n$

$\Rightarrow R(x) \equiv 0$.

Example: $q=7, n=5, k=3; d=3, e \leq 1$

$$RS(F(1,2,3,4,5); 5; 3), u = (u_0, u_1, u_3) = (1, 1)$$

$$f_u = 1+x+x^2$$

Sent Codeword: $(3, 0, 6, 0, 3)$
 $\downarrow$ channel

Received word: $(3, 1, 6, 0, 3)$

$E(x), Q(x)$

$$\uparrow$$

 $x+b_0$

$$\uparrow$$

$$a_0 + a_1 x + a_2 x^2 + a_3 x^3$$

$$1 \rightarrow 3(1+b_0) = (a_0 + a_1 + a_2 + a_3)$$

$$2 \rightarrow 1(2+b_0) = (a_0 + 2a_1 + 4a_2 + a_3)$$

$$3 \rightarrow 6(3+b_0) = (a_0 + 3a_1 + 2a_2 + 6a_3)$$

$$4 \rightarrow 0(4+b_0) = (a_0 + 4a_1 + 2a_2 + a_3)$$

$$5 \rightarrow 3(5+b_0) = (\dots)$$

Upon solving: $a_3 = 1, a_2 = 6, a_1 = 6, a_0 = 5, b_0 = 5$